

ECSE 4560/6560: MCS Project Topics

ECSE 6560 students (and research-oriented ECSE 4560 students):

Matlab: You can explore one of the following Matlab toolboxes that you can use to design and run some experiments.

- LTE communications
- Bluetooth communications
- WLAN communications
- 5G (including mmWave) communications
- Multi-antenna and MIMO communications
- AI for wireless communications
- Satellite communications

Hardware: You can utilize hardware/devices in MCS/IoT lab such as PlutoSDR, RFID, mmWave Radar, mmWave array

Dataset: You can take any one of the datasets in the following NSFs-sponsored project data repositories and analyze them (A slide set summarizing two of the datasets, from Northeastern and Arizona State are attached).

- Northeastern: <https://www.rfdatafactory.com/datasets>
- ASU: <https://www.deepsense6g.net/scenarios/>
- Columbia: <https://wiki.cosmos-lab.org/wiki/Datasets#Datasets>
- NCSU: <https://aerpaw.org/experiments/datasets/>
- UCSB: <https://web.ece.ucsb.edu/mostofi-lab/code-data.html>
- UCSD: <https://wcsng.ucsd.edu/code/>
- POWDER <https://powderwireless.net>
- NIST <https://nextg.nist.gov/submissions/list>

These are some classic or fundamental papers that introduces/advances topics in wireless communication and wireless sensing. Feel free to suggest any other papers not in the list and discuss with the instructor.

Localization

- ArrayTrack: A Fine-Grained Indoor Location System [USENIX NSDI 2013]
<https://www.usenix.org/conference/nsdi13/technical-sessions/presentation/xiong>
- SpotFi: Decimeter Level Localization Using WiFi [ACM SIGCOMM 2015]
<https://dl.acm.org/doi/10.1145/2785956.2787487>
- Chronos: Decimeter-Level Localization with a Single WiFi Access Point [NSDI 2016]
<https://www.usenix.org/conference/nsdi16/technical-sessions/presentation/vasisht>
- mD-Track: Leveraging Multi-Dimensionality for Passive Indoor Wi-Fi Tracking [Mobicom 2019]
<https://dl.acm.org/doi/10.1145/3300061.3300133>

MIMO & Distributed MIMO

- JMB: Scaling Wireless Capacity with User Demands [ACM SIGCOMM 2012]
<https://courses.grainger.illinois.edu/ece598hh/fa2020/papers/megamimo.pdf>
- MegaMIMO: Real-time Distributed MIMO Systems [ACM SIGCOMM 2016]
<https://courses.grainger.illinois.edu/ece598hh/fa2020/papers/megamimo2.pdf>
- IAC: Interference Alignment and Cancellation [ACM SIGCOMM 2009]
<http://ccr.sigcomm.org/online/files/p159.pdf>
- AirShare: Distributed coherent transmission made seamless [IEEE Infocom 2015]
<https://ieeexplore.ieee.org/abstract/document/7218555>

Millimeter-Wave & 5G

- Fast Millimeter Wave Beam Alignment [Sigcomm 2018]
<https://synrg.csl.illinois.edu/papers/agile-sigcomm18.pdf>
- Multi-Stream Beam-Training for mmWave MIMO Networks [Mobicom 2018]
<https://dl.acm.org/doi/10.1145/3241539.3241556>
- mmReliable: Two beams are better than one: Towards Reliable and High Throughput mmWave Links [Sigcomm 2021]
<https://wcsng.ucsd.edu/mmreliable/>
- mmFlexible: Flexible Directional Frequency Multiplexing for Multi-user mmWave Networks [Infocom 2023]
<https://wcsng.ucsd.edu/mmflexible/>

Smart Surfaces

- ScatterMIMO: Enabling Virtual MIMO with Smart Surfaces [ACM Mobicom 2020]
<https://wicsng.ucsd.edu/scattermimo/>
- LAIA: Towards programming the radio environment with large arrays of inexpensive antennas [NSDI 2019]
<https://dl.acm.org/doi/10.5555/3323234.3323259>
- RFocus: Beamforming Using Thousands of Passive Antennas [ACM NSDI 2020]
<https://people.csail.mit.edu/venkatar/rfocus.html>

Wireless Sensing & RF Imaging

- WiVi: See Through Walls with WiFi [ACM SIGCOMM 2013]
<https://people.csail.mit.edu/fadel/papers/wivi-paper.pdf>
- WiTrack: 3D Tracking via Body Radio Reflections [USENIX NSDI 2014]
<https://www.usenix.org/conference/nsdi14/technical-sessions/presentation/adib>
- WiTrack2: Multi-Person Localization via RF Body Reflections [USENIX NSDI 2015]
<https://www.usenix.org/conference/nsdi15/technical-sessions/presentation/adib>
- VitalRadio: Smart Homes that Monitor Breathing and Heart Rate [CHI 2015]
<https://www.mit.edu/~fadel/papers/vitalradio-paper.pdf>
- RFCapture: Capturing the Human Figure Through a Wall [ACM SIGGRAPH 2015]
<https://people.csail.mit.edu/hongzi/content/publications/RFCapture-TOG.pdf>
- Widar: Decimeter-Level Passive Tracking via Velocity Monitoring with Commodity Wi-Fi [Mobihoc 2017]
<https://dl.acm.org/doi/pdf/10.1145/3084041.3084067>

Machine Learning for RF Sensing

- EQ-Radio: Emotion Recognition Using Wireless Signals [ACM MobiCom 2016]
<https://eqradio.csail.mit.edu/files/eqradio-paper.pdf>
- RF-Pose3D: RF-Based 3D Skeletons [ACM Sigcomm 2018]
<https://rfpose3d.csail.mit.edu/>
- RF-Pose2D: Through-Wall Human Pose Estimation Using RF Signals [CVPR 2018]
<https://rfpose.csail.mit.edu/>

Millimeter-Wave Imaging & Sensing

- HawkEye: Through Fog High Resolution Imaging Using Millimeter Wave Radar [CVPR 2020] <https://jguan.page/HawkEye/>
- 3D Point Cloud Generation with Millimeter-Wave Radar [IMWUT 2020] <https://dl.acm.org/doi/abs/10.1145/3432221>

Security & Privacy

- mmSpy: Spying Phone Calls using mmWave Radars [IEEE S&P 2020] <https://www.cse.psu.edu/~mkg31/papers/mmSpy.pdf>
- MadRadar: A Black-Box Physical Layer Attack Framework on mmWave Automotive FMCW Radars [NDSS2024] https://www.ndss-symposium.org/wp-content/uploads/ndss2024_f153_paper.pdf
- Toward Spoofing-Resilient and Communication-Integrated MmWave Radar Sensing [Mobisys 2025] <http://www.phpathak.com/files/scr-mobisys.pdf>